

# A Reinforcement Learning-Based Dynamic Adaptive Gamification (DAG) Method for Online Employee Training

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**Abstract**— In times where working remotely and online work environments are on the rise; the traditional employee training method struggles to stay motivated and engaged. Gamification has emerged as a widely used motivational approach that can be tailored to user preferences to improve user engagement. While various machine learning techniques have been applied to gamification, the use of reinforcement learning for real-time adaptive personalization in employee training contexts remains comparatively underutilized. Therefore, this study presents a Reinforcement Learning-Based Dynamic Adaptive Gamification (DAG) method designed to enhance online employee training through real-time adaptation and personalization. The proposed approach leverages data-driven decision making, iterative feedback mechanisms, and personalization algorithms to optimize the training experience. The methodology comprises three key phases: (A) System design, (B) Implementation consisting the Adaptation and Personalization processes such as game interaction, ongoing feedback and (C) Evaluation. The results indicate the effectiveness of the DAG method in improving user engagement and increasing user retention in online training environment.

**Keywords**—Gamification, Reinforcement learning, Adaptive, Personalization, Engagement

## I. INTRODUCTION

Gamification has transpired into a widely adopted strategy to increase engagement and inspiration in online learning environments [1]. The inclusion of awards in game elements such as leaderboard, reward, mark, challenge, and non-game references promoted immersive and interactive experiences. [2]. Research has indicated that gamification can significantly improve learner engagement and knowledge retention [3]. Nonetheless, traditional gamification models typically rely on a well-constructed outline, where predefined game elements are equally appealed to all users, disregarding the user's engagement patterns and learning preferences [4].

Most of the adaptive gamification methods use limited classification models and fixed rule-based systems. These systems do not support real-time personalization or quick changes [5]. In response to this problem, a method named Dynamic Adaptive Gamification (DAG) is introduced in this study. DAG uses real-time user data, simple decision-making rules, optimization steps, and repeated feedback. Unlike the traditional ones that solely rely on learning based on rewards, DAG adapts to game elements in real time based on individual performance, engagement and user behaviour. The learning path changes based on how well the user performs and how interested they are with the ongoing training process. This creates a learning experience that fits the user and grows over time. The personalized learning encourages the users to stay

motivated and leads to improved engagement in online employee training. The proposed method measures how effectively the users remain engaged and complete the training. In addition, the study examines the accuracy of the method in convergence of the shift in user's traits when interacting with game element. DAG was benchmarked with existing gamification models that relies on static rules and offers limited adaptabilities highlighting its ability in personalization and engagement. The results demonstrates that DAG significantly enhances personalised learning and improves the overall effectiveness of learning experience.

## II. LITERATURE REVIEW

### A. Gamification

Traditional gamification models mostly use fixed frameworks. They follow a one-size-fits-all approach. These models use a predefined set of game elements and are based on the idea that user preferences and behaviour stay the same over time [6]. These assumptions are being questioned more frequently by various studies that show user experience which includes engagement and learning preferences are constantly evolving and differs greatly depending on a specific situation [7].

In response to these limitations, recent research has explored personalized, dynamic gamification, adapting game mechanics in real-time based on user behaviour and feedback. Unlike static models, adaptive gamification models are seen to continuously assess user behaviour and update game element in real-time to enhance motivation and optimize learning outcomes [8]. In an adaptive mechanisms comprising user feedback loops to dynamically modify game elements, the approach lacked a systematic framework for the continuous evolution of user preferences [9]. Similarly, another researcher examined the consequences of various game elements, highlighting the importance of tailoring experience for adaptive mechanisms that customize to individual user differences in motivation and engagement [10]. In another study, a player typology system to classify users based on interaction styles in gamified environments, suggests that different user types react differently to various game elements [11]. In a recent research, it was concluded that integrating machine learning techniques into gamification systems to provide real-time content adaptation, reveals the potential of AI-driven gamification strategies [12].

### B. Adaptive Gamification and Reinforcement Learning

Dynamic adaptation in gamification has been explored across various domains including corporate training, higher

education, and online learning platforms [13]. Recent study proposed an adaptive gamification framework that personalizes the learning experience based on real-time behavioural data, demonstrating significant improvements in user engagement [14]. It was further advanced by presenting a taxonomy of adaptive gamification techniques and classification based on the types of adaptive mechanism deployed and the degree of personalization [15]. The proposed taxonomy exhibits five dimensional structure associated to the learners and their learning environment that is employed to evaluate gamification design in learning environment[16]. Despite these advancements, most existing adaptive gamification methods rely on predefined adaptation rules or simplistic user classification models, limiting their ability to evolve dynamically based on complex and continuous user interactions [17].

The research done on gamification and its features has been driven with the aid of player modelling, contributing to better understanding of users' preferences and behaviour patterns. Bartles's Player Types Framework was one of the earliest developed frameworks distinguishing players into four main categories viz. Achiever, Explorer, Socialize, and Killer, depending on the type of motivation driving them in the gaming world [18]. This classification has been built on later by other scholars [19] using key motivational factors that guide user participation. Further development with real-time behaviour changes to player modelling that made it on-the-fly responsive to the elements of gamification showed the importance of continuous change adaptation systems [20]. Still, a big part of the existing adaptive approaches to gamification relies on rigid models of the users and fixated rules for changing the system, thus failing to keep up with the changing trends of user participation [21].

The use of Artificial Intelligence (AI) and gamification decision making systems have been studied in the past few years as a potential solution to adaptive gamification challenges [22]. The user adaptive level gamification interface functioning on Reinforcement Learning theory is designed to iteratively adjust the game mechanics based on users' interactions to enhance user immersion and optimize learning objectives [23]. Some studies have attempted to utilize reinforcement-based adaptation methods; however, these approaches often target specific elements of a game and not a holistic gamification framework that adapts and learns based on the evolving needs of the user [24]. The proposed study builds on previous research and proposes a reinforcement-based learning type dynamic adaptive gamification that will dynamically update game elements based on user interaction in the moment. The proposed method uses elements of optimization algorithms, utility-based decisions and provides iterative feedback to generate an evolving environment with a highly personalized experience. This study will also address the limitations of current static and rule-based adaptive gamification approaches offered due to the variety of available dynamic adaptation gamification systems to improve gamification approaches in online training environments [25].

### III. METHODOLOGY

The present study employed a structured methodology to develop and evaluate the Dynamic Adaptive Gamification (DAG) method for online employee training. The research methodology consists of three key phases: system design, implementation and evaluation.

#### A. System Design

The DAG method was built using a reinforcement learning system that dynamically adapts game elements in response to user interactions. The system uses simple decision-making rules, repeated feedback, and optimization steps to allow real-time changes for each user. The design steps included setting up basic game features, ways to understand users, and how to adjust the system.

A gamified training prototype was run in a simulated environment where users worked with learning modules. The system tracked real-time data on user engagement, preferences and performance. This data was used to update user profiles and change the game elements. The backend was created using Python and TensorFlow to support the learning system. The user interface was made with JavaScript and React. Flask and REST APIs handled the link between the frontend and backend. All user data and activity logs were saved in a PostgreSQL database. The system was run on AWS EC2 so it could support more users if needed. To support adaptive gamification, the system incorporated various game elements designed to enhance user motivation and engagement. These included points, badges, leaderboards, achievements/unlockable, progress bars, levels, challenges, avatars, and real-time feedback. While all users interacted with the same elements, only the DAGG group experienced dynamic, real-time personalization. Each game element was mapped to specific player traits based on the Hexad framework (e.g., Achiever, Free Spirit, Socializer) [2].

#### B. The DAG method implementation

The DAG method was implemented using a modular architecture that integrates various adaptation and personalization algorithms. The system captures interaction data in real time, updates player profiles dynamically, and adjusts the game mechanics accordingly. The DAG method implementation is structured into two primary phases: the Adaptation Phase and Personalization Phase.

##### 1) Adaptation Phase

**Player Initialization:** The users are required to configure their profiles which involves gathering data on player traits by using selection and setup algorithms. The selection and setup algorithms are crucial for determining the right game adaptation which leads to personalized experience. It consists of processes to gather player data, setup the players profile as follows:

*update\_player\_ratings(player, selected\_game\_element, e)* (1)

Based on the selected game element and learning rate, the function updates the players profile. However, the detest player is treated differently as they have no initial information. If the trait is dominant, the value will be reassigned to prevent negative game experience. According to the given data and to configure the game environment for a personalized interaction. Weighted average is used to update the players trait vector:

*new\_value = (1 - e) \* old\_value + e \* interaction\_value* (2)

**Game Interaction:** This phase consists of activities such as the selection of game elements, processing interaction data, updating player traits and utilities, and storing interaction history. Players interactions with gamification elements are processed through a utility-based decision-making model.

$np.random.choice(len(game\_elements), p=utilities)$  (3)

An interaction simulation takes place where a score based on the interaction is generated for each of the trait following the players current profile. The selected game returns an interaction value where:

**Feedback and Adjustment:** The player ratings are updated based on the interaction history. The interaction parameters are adjusted using optimization algorithm.

**Iteration and Monitoring:** The interaction history is used to continuously monitor player progression. Real time iterative updates are carried out to refine the player's experience.

The sub-activities within the Adaptation phase includes:

**Selection and setup modules:** for player initialization. This is to initialize the players experience prior to engagement with the system. This step involves gathering player information and to set up the gamification environment to allow the game element adaptation effectively.

**Utility-based decision engine:** for adaptive interactions. This is where the system decides on the best response to present to the player based on real-time data. The engine uses utility theory to prioritize decisions. Utility theory depicts players interactions with game element and evaluates game elements based on their personality type and previous interaction to make decision.

**Optimization algorithm:** for feedback-based adjustments. The optimization algorithm is responsible in refining game interactions based on players feedback to ensure the game element continuously evolves and adapt to the players preferences. It occurs through an iterative update of player traits based on players interaction with game element using the formula as follows to update the players traits.

$$(1 - e) * \text{value} + e * \text{interaction\_value} \quad (4)$$

## 2) Personalization Phase

**Customization:** This phase consists of customizing game experiences based on player preferences using user-driven customization algorithms.

**Dynamic Adjustment and Ongoing Feedback:** Player traits and preferences were dynamically adjusted using real-time adaptation and feedback algorithms.

The sub-activity within the Personalization phase includes:

**User-driven customization system:** for a personalized experiences. Players can choose to interact with their preferred game element. The system allows the players to personalize their experience according to their preferences which contributes to enhanced engagement.

## C. Evaluation

The efficacy of the proposed Dynamic Adaptive Gamification (DAG) method was assessed using a controlled experimental setup involving participants undergoing online employee training. The evaluation was conducted using a custom-built online training platform. This platform was made to deliver gamified learning modules and track user activity. A total of 40 academic staff from a local university participated in the study. The evaluation lasted two weeks. Participants were randomly placed into two groups:

**Static Gamification Group (SGG):** These participants received a fixed set of game features. There was no personalization or real time adaptation.

**Dynamic Adaptive Gamification Group (DAGG):** These participants engaged in a personalized learning system that updated game elements dynamically in real time. The adaptation was based on each person's engagement, performance, and behavioural data collected during the training.

Both groups interacted with the same training content. The content focused on topics for professional growth for academic staff. It was divided into daily lessons. Each lesson included reading online materials, watching short videos, answering quizzes, and doing small tasks like multiple-choice or drag-and-drop questions. Prior to starting the training, DAGG participants answered the Hexad Player Type Questionnaire suggested by [26] and approved by [27]. This gave an early profile for each person (for example, Achiever, Socializer, Free Spirit). These profiles helped the DAG system set up game features at the start. The system used reinforcement learning to adjust the gamification in real time. The system tracked each user's behavior, such as time spent on tasks, quiz results, login frequency, and activity patterns. It updated player traits at each iteration. These updates informed ongoing adaptation of game elements, such as adjusted challenge levels, tailored feedback, and personalized rewards. Table I outlines the implementation differences of key game elements across the two groups.

Table I: Comparison of Game Element Implementation

| Game Element            | DAGG (Dynamic)                                       | SGG (Static)                           |
|-------------------------|------------------------------------------------------|----------------------------------------|
| Points                  | Adjusted based on player preferences                 | Fixed amount for all actions           |
| Badges                  | Awarded for behaviours matching player type          | Awarded after task completion          |
| Leaderboards            | Personalized views, segmented by group               | Single public leaderboard              |
| Challenges              | Tailored to player engagement patterns               | Same for all players                   |
| Feedback                | Real-time and tailored                               | Pre-set and generic                    |
| Avatars/Profiles        | Evolve based on player actions                       | Static avatars                         |
| Progress Bars           | Adaptive pacing and rewards                          | Linear tracking                        |
| Achievements/Unlockable | Dynamic, behaviour triggered, unlock special content | Static, limited, no unlocking function |

To evaluate the impact of DAG compared to traditional gamification approaches, the study employed multiple Key Performance Indicators (KPIs): User engagement, motivation and knowledge retention. These were assessed using detailed usage evaluation metrics explained in Table II. Additionally, a convergence analysis was conducted to assess the accuracy of the DAG method in tracking and adapting to dynamic changes in player traits over time. A comparative analysis between DAGG and SGG also helped evaluate the degree to which the DAG method outperformed static gamification models in maintaining and improving user engagement.

Table II: Evaluation metrics

| Metric                        | Description                                                                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average session length (mins) | Measures the average amount of time users spend during a single session in the gamification system.                                                                                             |
| Average interactions          | Indicates the average interactions of user with the system or game elements during a session which includes rating of game element.                                                             |
| Average completion rate (%)   | Refers to the percentage of users who complete a session or reach a certain goal within the system (such as finishing a game level, completing a task, or completing the learning process).     |
| Retention Rate (7 days)       | Measures the percentage of users who return to the system within a specified period of 7 days after their initial interaction. This metric measures the retention of long term user engagement. |

#### IV. RESULTS AND DISCUSSION

The experiment yielded notable differences between the DAGG and SGG groups across several performance and engagement dimensions.

##### A. Comparative evaluation between the SGG and DAGG groups.

###### 1) Average session length (mins)

The average session length varied significantly between the DAGG and SGG. The DAGG participants demonstrated longer session lengths, which suggests that users were engaged and interested in the gamification system for a longer period. In this case, the session length increased by 2.20 minutes indicating, that the dynamic adaptive system is likely to keep users engaged for more time in a way that suits individual user preferences.

###### 2) Average interactions

As shown in Table III, the average interaction rate was elevated in the DAGG group (20%) in comparison with the SGG group (17.8%), indicating significant user engagement in the DAGG group. The data indicate that users in the DAGG group interact more, which suggests that they are actively involved with the system rather than passively consuming content. The increase (+3 interactions) reflects that dynamic adjustments to the system are likely to spark more engagement, possibly because the gamification system is more personalized and offers more compelling interaction opportunities.

###### 3) Average completion rate (%)

A higher completion rate indicates that users are more likely to finish what they start, which is a strong indicator of user satisfaction and engagement. The DAGG group shows an increase in the completion rate (+4.20%), which suggests that the dynamic system's personalized elements make it more likely for users to stay on track and finish their tasks according to their preferences while the SGG group indicated an 83.4% completion rate.

###### 4) Retention Rate (7 days)

A +5% increase in the 7-day retention rate from the DAGG group means that more users are returning indicating the retention rates are significantly higher in the DAGG group

(46.5 %) than in the SGG group (41.5%), this possibly could be interpreted that users are returning, likely because the system's ability to adapt to user needs keeps them interested and engaged over a longer period of time and allows them to complete the learning process.

Table III: Results comparison of User engagement metrics

| Metric                        | DAGG  | SGG   | Difference | Insight                                                                                                              |
|-------------------------------|-------|-------|------------|----------------------------------------------------------------------------------------------------------------------|
| Average session length (mins) | 20.00 | 17.80 | +2.20      | DAGG users engage for longer sessions, likely due to dynamic adjustments to the gameplay.                            |
| Average interaction           | 12.60 | 9.60  | +3.00      | DAGG encourages more interactions potentially due to personalized content and dynamic game element changes.          |
| Average completion rate (%)   | 87.60 | 83.40 | +4.20      | DAGG results in a higher completion rate, suggesting better user satisfaction and engagement through personalization |
| Retention rate(7 days)        | 46.5  | 41.5  | +5.00      | DAGG adapts to player behavior in real time, helping sustain engagement.                                             |

##### B. Convergence accuracy analysis of the DAG method for player evolution

Figure 5 illustrates the evolution of all player trait values across six interaction iterations, with each line representing a specific trait (e.g., *Achiever*, *Player*, *Socializer*, *Free Spirit*). Initially, the user exhibits a dominant *Achiever* trait (value = 0.30), but as the DAG system adapts through real-time interaction data, other traits gradually increase. This progression reflects a shift from a single dominant trait toward a more balanced motivational profile, indicating behavioural convergence. The narrowing gap between the top and second-most dominant traits shows that the system is becoming more stable. This means it does not focus too much on early user behaviour. Instead, it keeps improving its predictions as more data is collected. Figure 6 shows the most dominant trait in each iteration highlighting the evolvement of the main trait from *Achiever* to *Free Spirit*. This change shows the system's ability to dynamically identify and react to changes in user motivation. The smooth transition between traits and the narrowing trait gap demonstrates the reliability of the DAG system's adaptation. The steady change in traits across iterations also shows that the algorithm doesn't converge randomly but through an informed utility-based adjustments. Concurrently, both the graph visualizations depict the convergence accuracy of the DAG method in exhibiting the user trait diversification and a reliable adaptation path over time hence supporting the empirical validation on the system's ability to produce a stable and personalized user models that evolves significantly with a continuous sustained interaction.

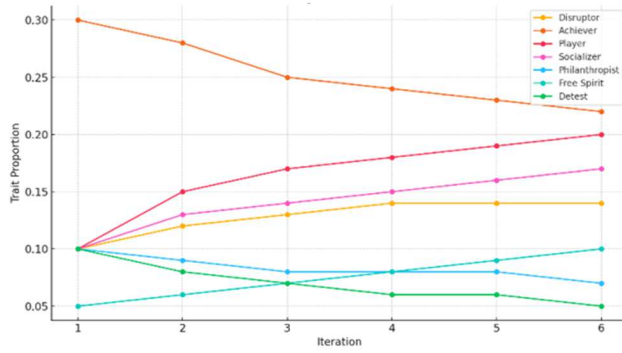


Fig. 5. Evolution of player traits with game element interactions

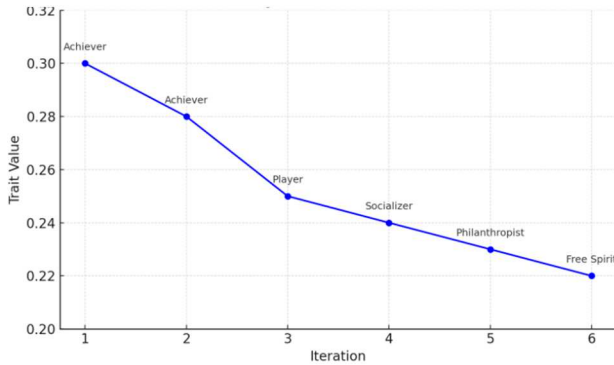


Fig. 6. Dominant Player Trait Value Over Iterations

### C. Comparative Analysis with Existing Approaches

As shown in Table IV, Comparative analysis revealed that DAG offers notable improvements.

Table IV: Comparative analysis of DAG method with existing work

| Feature                     | DAG method                                      | Lopes et al. (2019)                           | Klock et al. (2020)                                          |
|-----------------------------|-------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------|
| Real-time personalization   | Adjust content instantly based on user data.    | Static adjustments based on rules.            | Dynamic, but not real-time.                                  |
| Data-driven decision making | Use machine learning and predictive algorithms. | Rule-based, limited data use.                 | Uses analytics, but not predictive.                          |
| Sustained motivation        | Dynamically adjusts rewards to avoid fatigue.   | Can lead to repetitive or unbalanced rewards. | Changes challenges but may not sustain long-term.            |
| Engagement optimization     | Personalizes game experience in real-time.      | Adapt content but not fully personalized.     | Uses feedback loops but may miss the optimal learning curve. |

The performance of the proposed DAG method was benchmarked against two existing adaptive gamification frameworks from [28][29], with a focus on key performance indicators such as real-time personalization, data-driven decision-making, sustained motivation, and engagement optimization. Results indicate that DAG's use of real-time data to personalize game elements leads to higher levels of engagement, motivation, and knowledge retention. These enhancements position DAG as a viable and effective solution for corporate training contexts. The method enables

personalized learner experiences; dynamically adjusts learning paths based on individual progress, reduces dropout rates, and supports long-term skill development. Overall, DAG outperformed both static gamification and earlier adaptive models, offering superior responsiveness and personalization capabilities.

## V. CONCLUSION

The Dynamic Adaptive Gamification (DAG) method, a reinforcement learning-based adaptive method for online employee training, significantly enhances user engagement and increase user retention by adjusting game elements dynamically in real time adaptability. DAG is expected to provide a more personalized and interactive online training experience compared to the traditional existing static gamification models. It is leveraged with the integration of utility-based decision-making, optimization algorithms and iterative feedback loops that has enabled seamless adaptation based on user behaviour, addressing the limitations of previous gamification approaches effectively. These findings support the hypothesis that real-time adaptive gamification improves the effectiveness of training. The reinforcement learning based adaptation mechanism successfully classifies user preferences and adjusts gamification elements accordingly, leading to enhanced user engagement. Despite the observed improvements, some challenges remain, such as the need for additional onboarding and user familiarity with dynamic game mechanics. Future research should focus on expanding the DAG method to accommodate diverse learning environments and broader user demographics [30]. A possible direction is the integration of the hybrid adaptive strategies that combines user-driven customization with automated real-time adjustments.

The DAG method creates a personalised learning experience where the user does not only have control but is also able to benefit from the responsive smart platform which aims to boost engagement and learning outcomes. Further studies could explore to utilize advanced machine learning techniques such as Deep Reinforcement Learning (DLR) system that can be useful to personalize learning path dynamically and to handle complex user behaviour's while improving decision over time with smart automation to enhance the adaptability and personalization [31]. Another direction for future work is the implementation of DAG in validating its effectiveness in different training domains such as healthcare and educational institutions. Intuitive visualizations of adaptation mechanisms and enhanced onboarding processes, will also be explored to facilitate seamless interaction with the dynamic system.

Addressing these future research directions may enable the proposed DAG method to evolve into a robust comprehensive adaptive gamification method that enhances engagement in multiple training and educational settings and not limited to only corporate learning. This method will be able to support diverse learning profiles adjusting to different learning objectives. Eventually, enhancing user engagement and improving learning effectiveness over various environment.

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